

EAST CRAZIES EXAMPLE SOILS SCOPE OF WORK

Date: 2026-05-07

Audience: WLG demonstration package
Purpose: show that the system can produce resource-area scopes of work for a new NEPA project from a structured project intake
Example project: East Crazy Inspiration Divide land exchange
Forest: Custer Gallatin National Forest
District: Bozeman Ranger District
Resource area: Soils

This is a planning and contracting support SOW for an early-stage project. It translates the controlling source stack into the fieldwork, analysis, and reporting needed for a defensible soils specialist report or approved issue-screening memo.

Project drivers: proposed trail construction or reroute, trailhead improvements, stream/wetland approaches, wet or hydric soils, erosion and sediment delivery, parcel restrictions, access/easement features, grazing-related improvements, and acquired-parcel management.

This SOW is not a final agency decision, legal sufficiency review, or NEPA compliance determination.

DELIVERABLES

1. Fieldwork

- Survey-trigger table tying each proposed action element to the controlling source that requires field verification.
- Fieldwork plan, maps, and activity-area definitions for triggered soils surveys.
- Field data package with GPS points/tracks/polygons, completed field forms, photo log, soil map unit checks, DSD observations, wetland/riparian interface notes, and data gaps.

2. Analysis

- Existing-condition summary by activity area.
- Effects indicators and soil-disturbance calculations for no action, proposed action, and any Forest Service-provided alternative.
- Forest Plan consistency analysis using Appendix A.
- Mitigation/BMP/monitoring matrix tied to action element, location, source trigger, timing, responsible party, and verification method.
- Cumulative-effects and uncertainty summary where a cause-and-effect relationship exists.

3. Reporting

- Draft Soils Specialist Report or approved issue-screening memo.
- Appendix A authority-to-report crosswalk updated for the project.
- NEPA-ready soils summary text for later disclosure, if requested by the Forest Service.
- Final report or memo, final PDF, editable source file, tables, map atlas, GIS package, photo log, field forms, source inventory, metadata/readme, and limitations memo.

FIELDWORK/DATA COLLECTION

Field surveys will collect existing soil condition data within the proposed project area to evaluate soil disturbance, stability, sensitivity, and productivity. Field monitoring should be prioritized in areas with higher past disturbance, sensitive soil map units, wet or hydric soils, unstable slopes, erosion or sediment-connectivity risk, and areas expected to receive higher levels of ground disturbance from trail, trailhead, access, crossing, staging, restoration, or parcel-management work.

Before mobilization, prepare a survey-trigger table that ties each proposed action element to the source that requires field verification. If a survey below is not triggered, record the reason in the table.

Methodology

Required fieldwork is based on R1 Soil Quality Standards and soil-management direction, NFMA and 36 CFR part 219 project consistency requirements, and the Custer Gallatin National Forest Land Management Plan. Use the Forest Service Soil-Disturbance Field Guide, Forest Soil Disturbance Monitoring Protocol, NRCS Web Soil Survey/SSURGO, and National BMP direction as the field and data methods.

- DSD surveys. Required where a management activity could create or add to detrimental soil disturbance in an activity area. Classify and estimate existing DSD and expected DSD for trail, trailhead, access/staging, restoration, crossing, and parcel-management work. Record compaction, rutting, displacement, puddling, surface erosion, severe burning if present, organic-matter loss, and mass movement. Tie results to FW-STD-SOIL-01 and MON-SOIL-01.
- Soil survey and activity-area delineation. Required for each ground-disturbing action area. Verify soil map units, slope class, landform, drainage position, hydric/wet soil indicators, erosion hazard, revegetation limits, compaction/rutting limits, and sediment connectivity. Define the activity area used for DSD evaluation; for linear trails, document length, width, acreage, and rationale.
- Soil sensitivity and productivity survey. Required where soil map units or field indicators show productivity risk, wetness, shallow or restrictive layers, erosion hazard, compaction sensitivity, poor revegetation potential, or other sensitive soil conditions. Identify design changes,

restoration needs, or monitoring needed to maintain soil productivity and function.

- Slope, stability, and equipment-constraint survey. Required where mechanized construction, ground-based equipment, temporary access, blading, scarification, pits/trenches, burn piles, steep slopes, or unstable landforms are proposed. Verify sustained grades over 40 percent, high mass-wasting potential, unstable slopes, wet soils, and locations where FW-GDL-SOIL-01 through FW-GDL-SOIL-08 are triggered.
- CWD and organic-substrate survey. Required where vegetation activity areas, burn piles, slash treatment, forest-floor removal, revegetation/restoration, or other work affects coarse woody debris, litter, duff, or organic soil cover. Estimate percent coarse woody debris where MON-SOIL-02 is triggered and document restoration substrate conditions.
- Waterbody interface and BMP survey. Required where trail, trailhead, access, crossing, drainage, or restoration work is near streams, wetlands, riparian areas, seeps, springs, swales, drainage paths, or other sediment-routing features. Identify drainage controls, hydrologic disconnection, crossing hardening, wet-weather restrictions, revegetation, decompaction, and inspection needs.

Minimum Survey Record

For each triggered survey location or activity area, record:

- date, observer, weather, soil moisture, access limits, survey trigger, and source driver;
- activity-area ID, action element, soil map unit, GPS identifier, photo IDs, and field form ID;
- protocol used, measurements taken, disturbance class, percent disturbance where required, and linear-feature substitute metric if approved;
- slope, landform, drainage position, hydric/wet soil indicators, unstable-area indicators, productivity or sensitivity indicators, and sediment-connectivity notes;
- BMP, design criterion, mitigation, restoration, monitoring requirement, data gap, or revisit need.

Conduct fieldwork during snow-free conditions when the mineral soil surface, organic cover, and drainage paths can be observed. Conduct a wet-season or post-storm revisit only if waterbody connectivity, runoff, erosion, or sediment delivery cannot be determined during the main survey.

ANALYSIS OUTPUT

Use Appendix A as the authority-to-report control table. The analysis should answer only the soil questions triggered by the proposed action and should tie each conclusion to field evidence, GIS data, cited source direction, or a documented Forest Service assumption.

1. Existing Condition

Summarize the soil setting for each affected activity area:

- soil map unit, slope, landform, drainage, hydric/wet soil indicator, erosion hazard, and

- revegetation or compaction limitation;
- existing detrimental soil disturbance, route drainage, erosion, sediment connectivity, and restoration need;
- CWD, litter/duff, ground cover, or organic-substrate condition only where the CWD/organic-substrate survey is triggered;
- disposal-parcel and acquisition-parcel soil conditions only where parcel restrictions or future NFS management create a soil issue;
- data gaps that limit effects conclusions.

2. Effects Indicators

Use the same indicators for no action, proposed action, and any Forest Service-provided alternative:

- activity-area acreage for area-based work;
- length, width, crossing count, drainage feature count, wetland/riparian approach count, and representative disturbance class for linear trail work;
- percent detrimental soil disturbance where FW-STD-SOIL-01 applies;
- hydrologic connectivity, erosion risk, revegetation feasibility, wet-soil limitation, unstable-area risk, and restoration need.

Do not force linear trail effects into acreage-only conclusions. Use acreage where it is useful and linear metrics where they better describe the effect.

3. Proposed-Action Effects

Analyze soil effects from the proposed trail construction, trail reroute, trailhead work, crossings, access/staging, restoration, and parcel-management changes. Address whether the action would:

- create, add to, or restore detrimental soil disturbance;
- route sediment toward streams, wetlands, riparian areas, seeps, springs, swales, or drainage paths;
- reduce an existing stream/wetland crossing effect through the Inspiration Divide Trail reroute;
- require erosion, compaction, drainage, topsoil, revegetation, wet-weather, or equipment controls;
- create soil-management obligations on acquired NFS parcels or reduce soil risk through parcel restrictions after conveyance.

4. Forest Plan Consistency

Use Appendix A to document the Forest Plan consistency finding needed for the soils report. The report must state:

- whether FW-STD-SOIL-01 applies, is met, requires restoration, or is not triggered;
- which FW-GDL-SOIL-01 through FW-GDL-SOIL-08 guidelines apply and what design feature or equivalent

measure satisfies each one;

- whether MON-SOIL-01 or MON-SOIL-02 monitoring is required or screened out;
- which related road/trail, watershed, riparian, wetland, revegetation, invasive-species, backcountry, or recommended-wilderness components change the soils analysis.

5. Mitigation, BMPs, And Monitoring

Provide one mitigation/BMP/monitoring matrix. Each row must include action element, location, triggering source, soil concern, required measure, timing, responsible party, and verification method. Include only measures tied to a triggered source or a site-specific field finding.

6. Cumulative Effects And Uncertainty

Evaluate cumulative soil effects only where a cause-and-effect relationship exists. Use the same indicators listed above and consider existing/proposed routes, trailheads, unauthorized routes, drainage, grazing, wetland/riparian disturbance, foreseeable trail or road work, private land development risk after conveyance, and vegetation/restoration actions.

Close the analysis with unresolved assumptions, field limitations, missing design details, and Forest Service decisions needed before the soils conclusions can support later NEPA or implementation files.

REPORTING OUTPUT

1. Draft Soils Specialist Report Or Issue-Screening Memo

Provide a draft report or Forest Service-approved issue-screening memo with:

- project action elements and soils issue triggers;
- field methods and survey-trigger table;
- existing condition by activity area;
- effects indicators and disturbance calculations;
- direct, indirect, and cumulative effects where triggered;
- Forest Plan consistency finding using Appendix A;
- mitigation/BMP/monitoring matrix;
- residual uncertainty, assumptions, and limitations.

Acceptance criteria:

- directly addresses each soil-triggering action element;
- states whether FW-STD-SOIL-01 applies, is met, requires restoration, or is not triggered;
- ties conclusions to Appendix A, field evidence, GIS evidence, official soil data, or specialist judgment;

- distinguishes required mitigation or monitoring from optional design recommendations.

2. NEPA-Ready Soils Summary Text

If requested by the Forest Service, provide concise disclosure text drawn from the soils report or issue-screening memo.

Acceptance criteria:

- summarizes affected environment, environmental consequences, mitigation/design criteria, Forest Plan consistency, and cumulative effects only where supported by the report;
- references the soils report or issue-screening memo instead of duplicating technical detail;
- identifies design criteria or monitoring that must carry into later decision or implementation files.

3. Final Report And Data Package

Provide final report or memo, final PDF, editable source file, Appendix A, final tables, map atlas, GIS layers or geodatabase/geopackage, photo log, field forms, source data inventory, metadata/readme, limitations memo, and comment-response log.

Acceptance criteria:

- resolves Forest Service and ID-team comments;
- preserves source citations, field-data traceability, and the authority-to-report crosswalk;
- documents residual uncertainty and remaining Forest Service decisions;
- final PDF starts with a valid '%PDF-' header.

APPENDIX A. AUTHORITY-TO-REPORT CROSSWALK

Use this appendix as the starting control table for the soils specialist report. Update it only if Forest Service direction, source currentness, or the proposed action changes.

A1. NEPA, USDA NEPA Procedures, And FSH 1909.15

- Source: NEPA; USDA NEPA procedures at 7 CFR part 1b; FSH 1909.15.
- Trigger: Federal land exchange, trail, trailhead, access, and parcel-management decisions that need soils effects support for an environmental assessment or issue-screening memo.
- Report output: concise affected environment, methods, direct and indirect effects, cumulative effects if useful, mitigation/monitoring, source citations, and data limitations.
- Acceptance test: each conclusion is traceable to reliable existing data, field evidence, or a documented Forest Service assumption; the report does not make the agency decision.

A2. Third-Party Or Consultant-Prepared Analysis

- Source: 7 CFR part 1b contractor/third-party preparation procedures and Forest Service direction.
- Trigger: WLG or a consultant prepares soils analysis for Forest Service review.
- Report output: editable report, field forms, GIS/map package, photo log, source inventory, assumptions, limitations, and comment-response log.
- Acceptance test: Forest Service can independently review the methods, source citations, field evidence, and assumptions before using the report.

A3. NFMA And 36 CFR Part 219

- Source: NFMA; 36 CFR part 219, especially project/activity consistency with the applicable Forest Plan.
- Trigger: Any NFS project or authorization after approval of the Custer Gallatin Forest Plan.
- Report output: Forest Plan consistency section that addresses applicable desired conditions, standards, guidelines, suitability, and monitoring components.
- Acceptance test: the report states whether the action complies with standards, follows guidelines or uses an equally effective design, and avoids foreclosing applicable desired conditions.

A4. Custer Gallatin Forest Plan Soil Desired Conditions

- Source: FW-DC-SOIL-01, FW-DC-SOIL-02, and FW-DC-SOIL-03.
- Trigger: Trail, trailhead, crossing, access, restoration, and parcel-management actions that could affect soil productivity, hydrologic function, erosion, or ground cover.
- Report output: existing-condition and effects discussion for soil productivity, erosion, sediment connectivity, infiltration, ground cover, and restoration feasibility.
- Acceptance test: the report explains how the action maintains or moves toward the soil desired conditions, or identifies the design changes needed.

A5. Custer Gallatin Forest Plan Soil Standard

- Source: FW-STD-SOIL-01.
- Trigger: Any management activity that could create or add to detrimental soil conditions in an activity area.
- Report output: activity-area delineation, existing DSD estimate, expected DSD estimate, linear-trail metric rationale if used, and restoration/design measures needed to avoid exceeding the standard.
- Acceptance test: the report states whether FW-STD-SOIL-01 applies, is met, requires restoration, or is not triggered.

A6. Custer Gallatin Forest Plan Soil Guidelines

- Source: FW-GDL-SOIL-01 through FW-GDL-SOIL-08.
- Trigger: Ground-based equipment, temporary access, blading, scarification, pits/trenches, burn piles, steep slopes, high mass-wasting potential, CWD, or restoration work.
- Report output: guideline applicability table with action element, location, field finding, design feature, mitigation, or equally effective alternative.
- Acceptance test: every triggered guideline is carried into the design criteria or explicitly screened out.

A7. Custer Gallatin Forest Plan Soil Monitoring

- Source: MON-SOIL-01 and MON-SOIL-02.
- Trigger: DSD monitoring where FW-STD-SOIL-01 applies; CWD monitoring where vegetation activity areas or organic-substrate effects are present.
- Report output: monitoring recommendation with location, metric, timing, responsible party, and success criterion.
- Acceptance test: the report identifies pre/post DSD monitoring and CWD monitoring where triggered, or states why they are not triggered.

A8. Region 1 Soil Management Direction

- Source: USDA Forest Service Manual, Region 1 Supplement, FSM 2550 Soil Management.
- Trigger: Forest Service management activity that could affect soil productivity or create detrimental soil disturbance.
- Report output: regional soil-quality method, DSD threshold logic, productivity-risk discussion, and restoration requirements.
- Acceptance test: regional soil direction is visible in the methods, effects indicators, and mitigation/monitoring matrix.

A9. Soil-Disturbance Field Guide And Monitoring Protocol

- Source: Forest Service Soil-Disturbance Field Guide and Forest Soil Disturbance Monitoring Protocol.
- Trigger: Any required DSD survey or soil-disturbance classification.
- Report output: disturbance classes, field forms, photo/GPS evidence, activity-area summary, and protocol deviations.
- Acceptance test: disturbance conclusions are repeatable from the field record.

A10. NRCS Web Soil Survey And SSURGO

- Source: NRCS Web Soil Survey and SSURGO.
- Trigger: Every mapped activity area and any parcel where soil limitations affect the proposed action.

- Report output: soil map unit/activity-area crosswalk, limitation ratings, hydric/wet soil indicators, erosion hazard, drainage, revegetation limits, and data limitations.
- Acceptance test: official soil data are cited and field-verified where the action creates a soil issue.

A11. National BMPs

- Source: Forest Service National Best Management Practices Program.
- Trigger: Trail, trailhead, access, crossing, drainage, or restoration work with water-quality or sediment-delivery risk.
- Report output: BMP matrix for drainage control, hydrologic disconnection, crossing hardening, sediment control, wet-weather restrictions, decompaction, revegetation, and inspection.
- Acceptance test: BMPs are tied to mapped locations and verification methods, not listed generically.

A12. East Crazies Example Action Elements

- Source: East Crazies structured intake assumptions.
- Trigger: Sweet Trunk Trail work, Inspiration Divide Trail reroute, Big Timber Canyon Trailhead improvements, stream/wetland approaches, access/easement features, parcel restrictions, grazing improvements, and acquired-parcel management needs.
- Report output: action-element by soil-concern matrix, field-survey trigger table, effects indicators, maps, mitigation/BMP/monitoring matrix, and issue-screening rationale if a full report is not required.
- Acceptance test: every soils conclusion is tied to an action element or explicitly screened out.

SOURCES CITED

- National Environmental Policy Act, 42 U.S.C. 4321 et seq.
- National Forest Management Act, 16 U.S.C. 1600 et seq.
- USDA NEPA procedures, 7 CFR part 1b:
<https://www.ecfr.gov/current/title-7/subtitle-A/part-1b>
- National Forest System land management planning, 36 CFR part 219:
<https://www.ecfr.gov/current/title-36/chapter-II/part-219>
- Forest Service Handbook (FSH) 1909.15, National Environmental Policy Act Handbook:
https://www.fs.usda.gov/cgi-bin/Directives/get_dirs/fsh?1909.15=
- USDA Forest Service Manual, Region 1 Supplement, FSM 2550 Soil Management.
- Custer Gallatin National Forest Land Management Plan:
<https://www.fs.usda.gov/r01/custergallatin/planning/forest-plan/custer-gallatin-land-management-plan-forest-plan-revision>
- Forest Service Soil Quality Monitoring resources:
<https://www.fs.usda.gov/soils/monitoring.shtml>
- Forest Service Soil-Disturbance Field Guide:
https://www.fs.usda.gov/t-d/php/library_card.php?p_num=0819+1815P

https://www.fs.usda.gov/rm/pubs_other/rmrs_2009_napper_c001.pdf

- NRCS Web Soil Survey and SSURGO:

<https://www.nrcs.usda.gov/resources/data-and-reports/web-soil-survey>

- Forest Service National Best Management Practices Program:

<https://www.fs.usda.gov/naturalresources/watershed/bmp.shtml>

- East Crazies example project intake assumptions: land exchange, proposed trail construction, proposed trail reroute, trailhead improvements, parcel restrictions, easements, wetland/riparian interfaces, grazing-related improvements, and acquired-parcel management needs.